National AI Hackathon 2026 by SchubKI

16-21 of October 2026 (16 – opening, 17-18 – work, 19-20 – evaluation, 21 - closing)

Constructor Labs

Challenge name

Camera-based localization on a race track

Challenge description

Camera-based localization for autonomous racing.

Build a system that localizes a full-size autonomous race car on a circuit using cameras only (no GNSS, no LiDAR). The system must estimate the car's position and heading on track from onboard camera streams alone, accurately enough to follow a racing line at speed. The challenge is to go beyond standard visual odometry and deliver localization that holds up under motion blur, changing light, low-texture asphalt and race pace.

We will provide datasets from autonomous race car collected on F1 Yas Marina circuit.

Challenge (input + details)

Task. At camera rate (10 Hz) estimate the car's 2D position (x, y) and heading (yaw) in the track coordinate frame. The dataset has no LiDAR, radar, GNSS or IMU data. The first pose of the run is given and may be used as the initial pose.

Input data (per run, one ROS 2 mcap file):

- Images from the car's 7 onboard cameras at 10 Hz, with intrinsic and extrinsic calibration. Each image has its own exposure time stamp.
- Track map of Yas Marina Circuit: left/right track borders and pit lane borders in a local metric frame (metres)
- Ground-truth pose (timestamp, x, y, yaw) from the car's state estimation (RTK GNSS, lidar map matching, IMU and odometry fusion, about 0.1 m / 0.2°), provided for the training run only

Dataset splits:

- Training run (1194 s, night session), with ground truth, for developing and tuning methods

- Validation: teams choose their own laps from the training run and evaluate them with the provided scoring script
- Hidden test run (not included the dataset), a separate file without ground truth (first pose only), used for final ranking. It may include higher speeds and different lighting conditions than the training data

Key difficulties: motion blur at race speed, low-texture asphalt, repetitive track features, strong lighting changes (sun, shadows, artificial light), fast yaw rates in corners and long straights with few visual cues.

Processing mode: solutions must be causal. The estimate for frame t may only use data up to time t, as it would on the real car. Offline or smoothed results may be shown in the presentation but are not ranked.

Minimal requirements

- Camera-based localization. Only the camera images, camera calibration and track borders may be used. Car state data (accelerations, speed, and other fields) can be used for camera-based localization. Ground-truth poses (positions and orientation only used for training and for validation). No LiDAR, radar data. Ground truth may not be used at inference, except the first pose of the run.
- Causal (online) processing, with no future frames used
- Poses (timestamp, x, y, yaw) at camera rate (10 Hz expected) for the hidden test run. At least 90 % of the 1 s intervals of the run must have a pose to be ranked.
- Submission of both the pose files in the specified CSV format and the full source code. Submissions without code are not ranked.
- Reproducible code: a repository with a README, dependency list and a single run script that produces the submission files from the raw dataset. It must run on a single-GPU workstation. The organizers may re-run the code to verify results.
- Pre-trained models and public external datasets are allowed but must be declared
- Measured average processing time per frame, reported with the hardware used

Expected output

- **Pose files:** one CSV for the test run with the columns timestamp_ns, x [m], y [m], yaw [rad], in the track coordinate frame of the dataset. timestamp_ns is the header stamp of the image or image set used for the pose

- **Source code (mandatory):** a Git repository link or archive with all code, trained model weights (or a download link) and a run script that reproduces the pose files
- **Validation results:** output of the provided scoring script on the team's validation laps
- **Presentation (max. 5 minutes, max. 6 slides):** approach, system diagram, results, runtime, failure cases and lessons learned
- **Visualization:** a short video or plot of the estimated trajectory against ground truth on the track map (validation laps)

Evaluation criteria

Quantitative (70%), computed automatically on the hidden test laps:

- Absolute Trajectory Error: position RMSE [m] (25%)
- Lateral error RMSE across the ground-truth heading, the most critical measure for following a racing line (20%)
- Heading error RMSE [deg] (10%)
- Robustness: share of poses with position error below 1 m (10%)
- Runtime: average processing time per frame, reported by the team. Real-time capability (at or below the camera frame period) scores in full. (5%)

Qualitative (30%), scored by the jury:

- Technical soundness and novelty of the approach (15%)
- Engineering quality and reproducibility of the code (10%)
- Presentation, including analysis of failure cases and limits (5%)

Rules:

- Each quantitative metric is normalised across teams (best = full points)
- Submissions with less than 90 % coverage (1 s intervals with at least one pose) are not ranked
- Ties are broken by lower lateral error
- Missing source code, results the submitted code does not reproduce, using ground truth at inference, or non-causal processing in the ranked result leads to disqualification
- Practical target for context: sub-metre lateral accuracy at race speed

Dataset links

- **Dataset download:** the link
- https://eu1-s3.virtuozzo.com/dev-auto-mobility-data/hackaton_2026/c0_2026-10-13_22-53-42_camloc.zip
- **Starter kit (scoring script, ground-truth export script, example code to read the data; calibration and track borders are inside the mcap file):** included in the dataset zip file (tools folder and README)
- **INTRO:** Constructor Labs - Camera-based localization on a race track - INTRO.pdf
- **Contents:** camera images, camera calibration, track borders, car state data (accelerations, speed, and other fields), ground-truth poses (positions and orientation only used for training and for validation). No LiDAR, radar data.
- **Data format and structure:** described in the README included with the dataset
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